Search Frictions and Welfare in a Market for Prominence: Theory and Experiment*

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Abstract

In online marketplaces, sellers pay platforms to make their products more visible and easier to inspect, creating a market for search prominence. I study how search friction affects the level and distribution of surplus in this market. In my model, two *ex ante* identical sellers bid for a prominent position in a second-price auction and then simultaneously post prices. The consumer observes both prices and learns her match value for the prominent seller's product for free. She then decides whether to pay a search cost to inspect the other, non-prominent seller's product. I derive equilibrium bids, price distributions, and search cutoffs, and test these predictions in a laboratory experiment with low and high search costs. I find that platform auction revenue changes little as search cost rises, despite the predicted increase. Total and consumer surplus decrease with search cost, but consumer surplus falls proportionally much less than predicted and exceeds equilibrium benchmarks in both treatments. Sellers jointly retain only about 20 percent of total surplus after paying for prominence. Further analysis of bidding, pricing, and search behavior explains these findings.

JEL Codes: C72, C91, D83

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1 Introduction

Online marketplaces offer a wide range of products, but these products are not displayed equally prominently. Some appear at the top of search results or receive more visible display space, while others require consumers to scroll through several pages of results. The FTC's complaint against Amazon states that "70% of Amazon shoppers do not click past the first Search Results Page."[1] These differences in visibility can affect which products consumers inspect and ultimately buy. Sellers therefore pay platforms to make their products easier to discover and inspect, creating a market for search prominence. Examples include Google search ads, Amazon Sponsored Products, and sponsored hotel listings on travel booking platforms. These payments generate substantial platform revenue. In 2025, Alphabet reported $224.5 billion in "Google Search & other" advertising revenue.[2]

Costly consumer search gives sellers a reason to compete for prominence. In physical markets, search can involve traveling between stores to obtain product information. Online shopping reduces the physical costs of search, but consumers still need time to learn about a product's characteristics and assess whether they suit their tastes. A more prominent product can therefore attract purchases from consumers who stop before considering other products. How much a seller is willing to pay for this advantage depends on the additional revenue it expects to earn relative to a less prominent position. This value is determined in the product market, where sellers compete through pricing and consumers decide which products to search and buy.

This paper studies how search cost affects the level and distribution of surplus in a market for prominence. I develop a model in which sellers compete for prominence through a second-price auction and then compete through pricing in the product market. The value of prominence is determined endogenously by sellers' pricing and consumers' search and purchase decisions. The auction elicits sellers' willingness to pay for prominence through their bids. I test predictions for total surplus and its division among consumers, sellers, and the platform in a laboratory experiment with financially incentivized participants. I vary search cost exogenously to examine its effects on these welfare outcomes. I find that consumers receive more surplus than predicted, while sellers jointly retain only about 20 percent of total surplus after paying for prominence. I use observed bidding, pricing, and search behavior to explain these and other findings.

The expansion of online commerce has made price information more accessible. Prices appear in product listings and on price comparison websites, allowing consumers to compare them before clicking through to examine individual products. Sellers can therefore influence consumers' search decisions through their pricing strategies. A low price at an already

[1] See the Federal Trade Commission's Second Amended Complaint against Amazon, paragraph 97.
[2] Alphabet Inc., 2025 Annual Report, "Google Search & other" advertising revenue.

searched seller can encourage purchase without further search. A low price at an unsearched seller, however, can encourage further search by increasing the expected gain from inspecting its product. Prices thus affect not only how the surplus from a transaction is divided between buyer and seller, but also whether consumers search further and which products they buy.

I capture these pricing and search incentives in a parsimonious model with two *ex-ante* identical sellers and a buyer with unit demand. Each seller offers one product that may be a good or bad match for the buyer. After the auction allocates prominence, the sellers simultaneously post prices. The buyer observes both prices and learns her match value for the prominent seller's product for free. She then decides whether to pay a search cost to inspect the non-prominent seller's product. At any point, she may buy a product she has already inspected or exit without purchasing.

I implement the model in a between-subjects laboratory experiment with low and high search costs. I choose the experimental parameters to generate distinct predictions across these treatments. The model predicts that 1) sellers' bids and platform auction revenue rise one for one with search cost, 2) higher search cost reduces total surplus, consumer surplus, and combined seller profits, 3) both sellers use mixed pricing strategies, with the prominent seller charging a higher price on average, 4) higher search cost raises the prominent seller's price and lowers the non-prominent seller's price, and 5) search follows a cutoff determined by the observed prices and the buyer's initial match, with further search becoming less frequent at the higher search cost.

The laboratory addresses several challenges to studying this market in the field. Consumers may interpret a high ranking as a personalized recommendation by the platform or regard a seller's willingness to pay for prominence as a signal of product quality. These inferences make it difficult to disentangle the effect of a search-cost advantage from the information conveyed by prominence. Obtaining exogenous variation in search cost presents a further challenge. Platforms' search design may respond to factors that researchers do not observe and that also affect welfare and behavior. Transaction data alone also do not directly reveal consumers' match values or search costs. In the laboratory, I vary search cost exogenously while holding match distributions and the other market rules fixed. Because match values and search costs are assigned in the experiment, I can also calculate each welfare component from observed search decisions and transactions.

Consumers receive more surplus than predicted in both treatments, even though total surplus is below its equilibrium benchmark. Prominent sellers charge less than predicted, benefiting consumers while reducing seller revenue per sale. After a good initial match, these lower prices can also make further search unattractive, saving consumers a search cost. High prices charged by non-prominent sellers, however, can discourage search after a bad initial match and prevent valuable purchases. Sellers earn less revenue than predicted and pay a

substantial share of it to the platform, leaving them with only about 20 percent of total surplus.

Higher search cost reduces total and consumer surplus, but consumer surplus falls proportionally much less than predicted. Average bids rise by much less than predicted, and losing bids barely change. Since the winning seller pays the losing bid, platform auction revenue remains almost unchanged. Easier search therefore benefits consumers and raises total surplus in the experiment without substantially reducing platform auction revenue. The model's predicted conflict between auction revenue and consumer welfare is much less pronounced in the data.

The next subsection discusses the related literature. Section 2 presents the theoretical framework. Section 3 describes the experimental design and hypotheses. Section 4 reports the welfare and behavioral results. Section 5 examines how consumer search responds to prices and the initial match. Section 6 concludes. The appendices contain proofs, welfare calculations, and experimental instructions.

1.1 Related literature

The paper builds on models in which search frictions affect price competition. Diamond (1971) shows how costly price information can sustain monopoly pricing. With differentiated products, search also allows consumers to find a better match, as in Wolinsky (1986) and Anderson and Renault (1999). When prices are observable before search, they can influence which products consumers search and whether search continues. Choi et al. (2018), Ding and Zhang (2018), and Obradovits and Plaickner (2023) study pricing under this information structure. The present model retains costly learning about product fit with publicly observed prices, but sellers acquire the prominent position through an auction. A low price by the non-prominent seller can induce continued search, but does not move that seller ahead of the initial product information supplied by prominence. This separates the acquisition of the position from the subsequent use of prices to influence search, while allowing both decisions to respond to the same friction.

Prominence and ordered-search models explain how an advantage in reaching consumers changes price competition. Armstrong et al. (2009), Zhou (2011), and Armstrong (2017) study pricing when consumers encounter firms in a particular order. Armstrong and Zhou (2011) examine how firms obtain prominence, including through payments to intermediaries and competition in advertised prices. These papers connect the value of being considered early to subsequent pricing and consumers' continuation decisions. Field evidence also supports distinguishing search from purchase. Using hotel-ranking variation, Ursu (2018) provides evidence for a search-cost channel through which rankings affect consumer search. My focus is how the cost of searching competing products affects the value of a purchased search advantage and the distribution of surplus. The experiment makes the difference in search

costs explicit and holds sellers' match distributions fixed, so the winning position does not signal better product quality or a better match.

Competition for prominence also connects the paper to sponsored-search auctions. Edelman et al. (2007) and Varian (2007) analyze auctions allocating advertising positions with different click volumes. Chen and He (2011) and Athey and Ellison (2011) incorporate consumer search and the information conveyed by paid rankings. The closest theoretical comparisons for endogenous position values are Xu et al. (2011) and Anderson and Renault (2026). Xu et al. (2011) derive position values from subsequent price competition, while Anderson and Renault (2026) show how incentives to move in the search order depend on competition from firms ahead of and behind them. I bring this connection into the laboratory by observing bids together with the pricing and search decisions that generate the position's return. The second-price auction implements competition for the search advantage, allowing the experiment to examine how its value responds to search cost.

Experimental studies of search and pricing provide the basis for examining the behavior that generates these position values. Davis and Holt (1996), Abrams et al. (2000), and Cason and Friedman (2003) test how search and information frictions affect prices and market performance. Brown et al. (2018) examine sellers' use of exploding offers to discourage further search when prices are visible. Their experiment shows why consumers' responses matter for the profitability of a strategy intended to prevent continued search. The present experiment introduces competition for prominence into a market with endogenous pricing and consumer search. Sellers learn their auction-assigned positions before pricing, buyers can return to a previously searched product, and the treatment varies the cost of searching the non-prominent seller's product. Observing the auction and product market together permits a welfare comparison that includes the payment for prominence, the prices consumers pay, and the search costs they incur.

2 Theoretical Framework

2.1 Model

There are two ex-ante identical sellers and one buyer with unit demand. Each seller offers one product at zero marginal cost. The buyer's match values are independent across sellers, with

$$v_k = \begin{cases} V & \text{with probability } q, \\ 0 & \text{with probability } 1-q, \end{cases} \qquad k = 1, 2, \tag{1}$$

where $V > 0$ and $0 < q < 1$. The cost of searching the second seller's product satisfies $0 < s < qV$. The timing, information structure, payoff functions, match distribution, and

search cost are common knowledge. Realized match values are not public information. Sellers do not observe them when bidding or pricing, and the buyer learns each product's match value only when she searches that product.

The game has three stages, summarized in Figure 1. Stage 1 is the market for prominence. Stages 2 and 3 form the product market.

Stage 1. Auction for prominence. Sellers simultaneously submit bids in a second-price auction. The higher bidder receives the top position and pays the losing bid, $b^{(2)}$. The other seller receives the second position and pays nothing. If bids are equal, each seller has probability one-half of receiving the top position. After the auction, subscripts identify positions rather than seller identities. Seller 1 is the top seller and seller 2 is the second seller.

Stage 2. Pricing. After learning their positions and the position payment, sellers simultaneously post prices $p_1, p_2 \geq 0$.

Stage 3. Search and purchase. The buyer observes both prices and searches the top seller's product for free, learning v_1. She may pay s to search the second seller's product and learn v_2. Before searching the second seller, she may buy from the top seller or exit. After searching the second seller, she may buy from either seller or exit. Returning to the top seller is free, and buying from the second seller requires search.

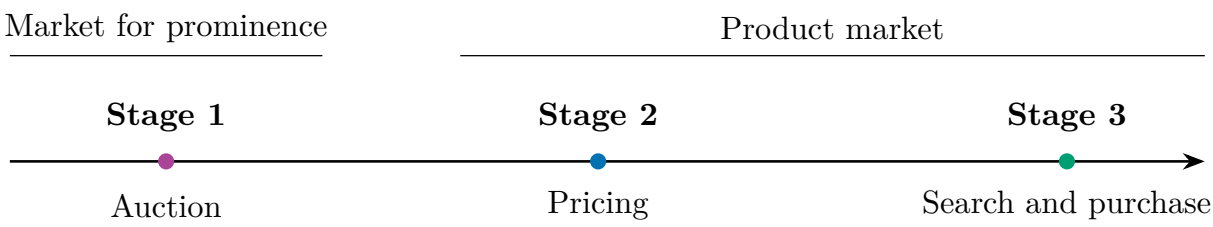


Figure 1: The three stages of the game

Let $x \in \{0, 1\}$ indicate whether the buyer searches the second seller and let $a \in \{0, 1, 2\}$ denote her purchase decision, with $a = 0$ denoting exit. The buyer's payoff is

$$u(a, x) = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \tag{2}$$

Seller revenues, seller profits, and platform revenue are

$$\Pi_k = p_k \cdot \mathbf{1}_{\{a=k\}}, \qquad \pi_1 = \Pi_1 - b^{(2)}, \qquad \pi_2 = \Pi_2, \qquad \text{PR} = b^{(2)}. \tag{3}$$

The position payment is sunk when sellers choose prices.

I solve the game backward. The buyer's search and purchase decisions determine each seller's demand at a given price pair. These demand functions yield a pricing equilibrium for the product market. The difference between the resulting expected revenues at the two positions determines the value of prominence and the sellers' auction bids.

2.2 Buyer search and seller demand

After observing (v_1, p_1, p_2), the buyer compares the expected gain from searching the second seller with its cost. With a bad top match, search offers a chance of finding a product worth buying. With a good top match, it offers a chance of buying a suitable product at a lower price. The following lemma specifies optimal choices, including choices at indifference.[3]

Lemma 1 (Optimal search cutoffs). *For $0 \leq p_1, p_2 \leq V$, the buyer's optimal search decision satisfies*

$$x^*(0, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_2 \leq V - \frac{s}{q}, \tag{4}$$

$$x^*(V, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_1 - p_2 > \frac{s}{q}. \tag{5}$$

Thus, at a search tie the buyer searches after a bad top match and stops after a good top match. She buys an available product with the largest nonnegative purchase surplus or exits if all available products have negative purchase surplus. At ties, purchase is preferred to exit and the top seller is preferred to the second seller.

Proof. See Appendix A.2, which also specifies optimal behavior after prices outside $[0, V]$.□

Figure 2 summarizes the buyer's choices. The complete buyer decision tree is in Appendix A.1.

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, let $D_k(p_1, p_2)$ be seller k's sale probability. The search cutoffs imply

	$D_1(p_1, p_2)$	$D_2(p_1, p_2)$
$p_1 - p_2 > \frac{s}{q}$	$q(1-q)$	q
$p_1 - p_2 \leq \frac{s}{q}$	q	$q(1-q)$.

(6)

When a buyer with a good top match stops, the top seller serves all such buyers, while the second seller sells only when the top match is bad and the second match is good. When the price gap induces search, the second seller also attracts buyers with two good matches. The top seller then sells only when the top match is good and the second match is bad. Figures 3 and 4 show the purchase paths in the two cases. At a zero price, the purchase tie rule can change unit demand, but the zero-price seller's revenue is always zero.

[3]The tie rules support the pricing equilibrium derived below. Appendix A.1 states them formally.

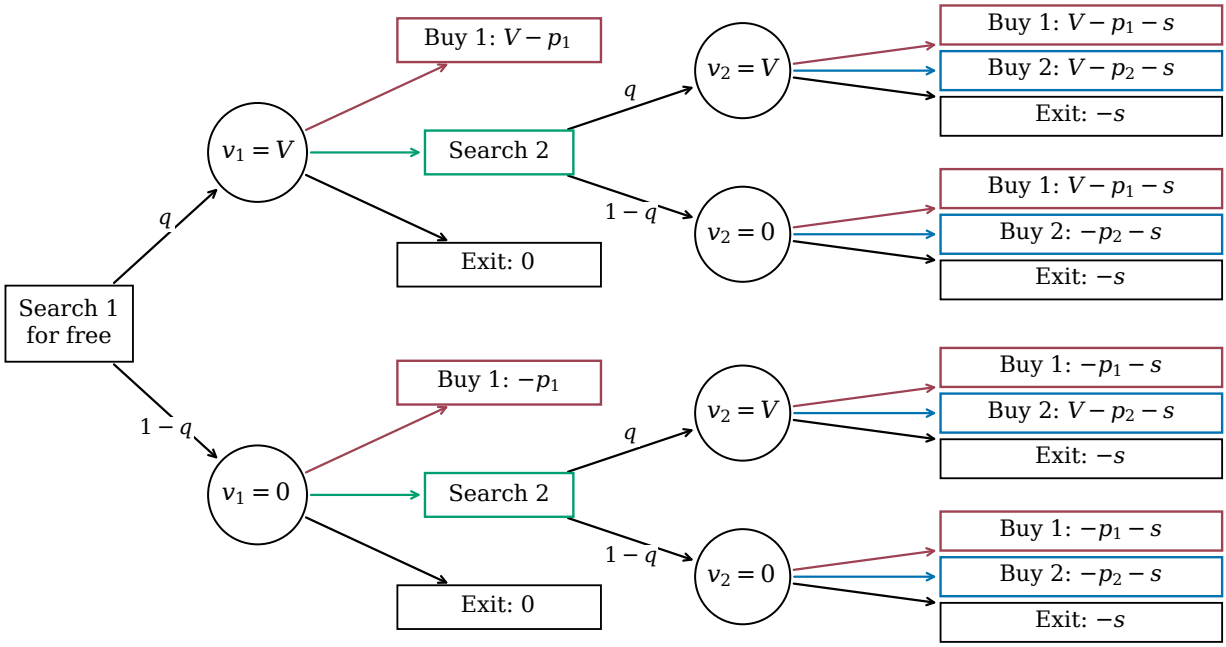


Figure 2: The buyer's search and purchase decisions

The buyer observes both prices and learns the top match for free. Search of the second seller costs s. Circles show realized match values and rectangles show choices.

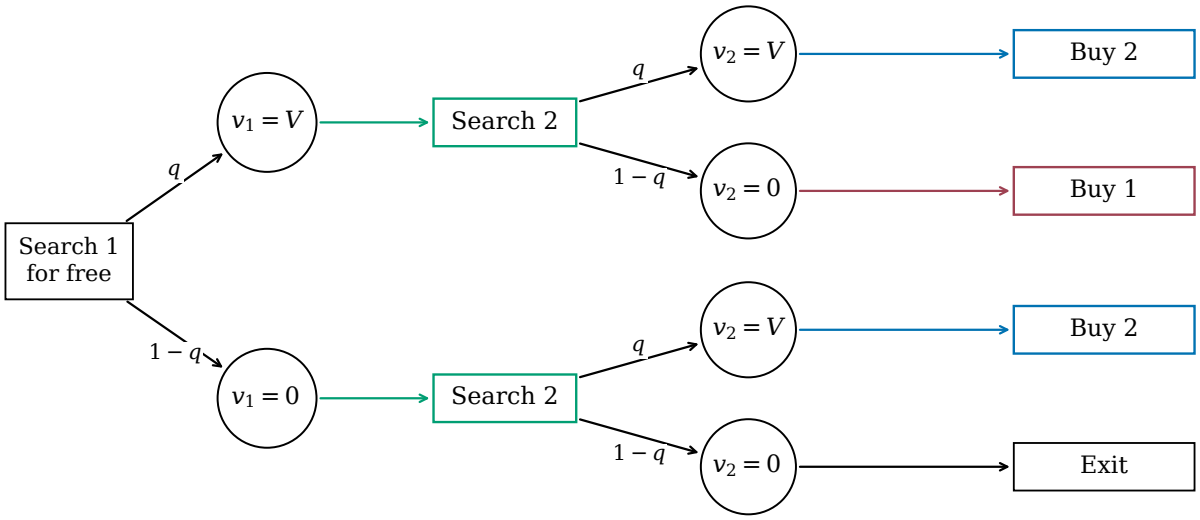


Figure 3: Search after a good top match when $p_1 - p_2 > \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the buyer searches the second seller after either top match.

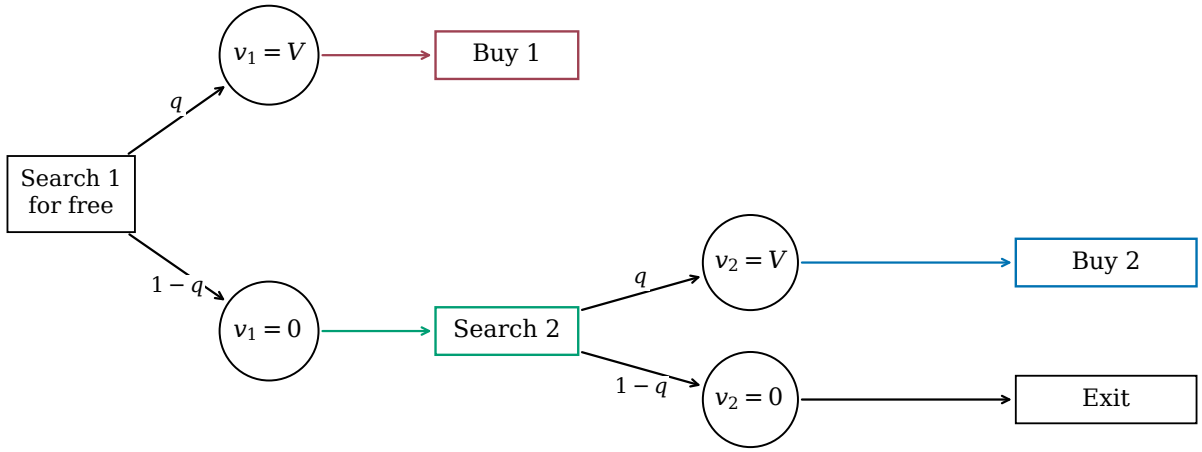


Figure 4: No search after a good top match when $p_1 - p_2 \leq \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the buyer stops after a good top match and searches the second seller after a bad top match.

2.3 Pricing equilibrium

Proposition 1 (No pure-strategy Nash equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the pricing subgame has no pure-strategy Nash equilibrium.*

Proof. See Appendix B.1. □

On a constant-demand segment, each seller benefits from raising its price. At the search cutoff, the second seller can instead gain demand by slightly lowering its price. These opposing incentives motivate mixed pricing strategies. At deviation prices outside the domain in (6), demand follows the buyer's optimal strategy.

Definition 1 (Mixed-strategy pricing equilibrium). We seek distribution functions $F_1(p_1)$ and $F_2(p_2)$ with connected supports $[\underline{p}_1, \overline{p}_1]$ and $[\underline{p}_2, \overline{p}_2]$ such that

$$p_1^* \in \arg\max_{p_1 \in [0,\infty)} p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)], \qquad \forall p_1^* \in [\underline{p}_1, \overline{p}_1], \tag{7}$$

$$p_2^* \in \arg\max_{p_2 \in [0,\infty)} p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)], \qquad \forall p_2^* \in [\underline{p}_2, \overline{p}_2]. \tag{8}$$

Theorem 1 (A mixed-strategy pricing equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the following independent distributions constitute an equilibrium*

of the pricing subgame:

$$F_1(p_1) = \frac{1}{q}\left[1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_1 - \frac{s}{q}}\right], \qquad \underline{p}_1 \le p_1 \le \overline{p}_1, \tag{9}$$

$$F_2(p_2) = \frac{1}{q}\left[1 - \frac{(1-q)V + s}{p_2 + \frac{s}{q}}\right], \qquad \underline{p}_2 \le p_2 < \overline{p}_2, \tag{10}$$

where

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \overline{p}_1 &= V, \\ \underline{p}_2 &= (1-q)\left(V - \frac{s}{q}\right), & \overline{p}_2 &= V - \frac{s}{q}. \end{aligned} \tag{11}$$

Both distributions are zero below their lower bounds and one at or above their upper bounds. The top seller's distribution is atomless. The second seller places mass $\frac{s}{qV}$ *at* $\overline{p}_2$.

Proof. Appendix B.2 verifies both optimality conditions against all nonnegative price deviations. □

The support bounds satisfy

$$\underline{p}_1 - \underline{p}_2 = \overline{p}_1 - \overline{p}_2 = \frac{s}{q}. \tag{12}$$

A higher search cost raises the lower bound of the top seller's price distribution and lowers both bounds of the second seller's distribution. To induce search, the second seller must offer a larger price reduction relative to the top seller.

2.4 Bidding for prominence

Theorem 2 (Symmetric bidding equilibrium). *Suppose the sellers use the pricing equilibrium in Theorem 1 after every auction outcome. Their expected revenues before any position payment are*

$$\mathbb{E}[\Pi_1] = q[(1-q)V + s], \qquad \mathbb{E}[\Pi_2] = q(1-q)\left(V - \frac{s}{q}\right). \tag{13}$$

The value of prominence is their difference,

$$\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] = s. \tag{14}$$

The symmetric bids $b_1 = b_2 = b^* = s$, together with these pricing strategies and optimal buyer behavior, constitute a subgame-perfect equilibrium. Platform revenue and expected seller profits are

$$\mathrm{PR} = s, \qquad \mathbb{E}[\pi_1] = \mathbb{E}[\pi_2] = (1-q)(qV - s). \tag{15}$$

Proof. See Appendix C.1. □

The auction payment absorbs the top seller's expected revenue advantage. A higher search cost raises the value of prominence and the equilibrium payment to the platform, while reducing both sellers' expected profits.

2.5 Welfare

The purchase decision is $a = 0$ for exit and $a = 1$ or $a = 2$ for purchase from the corresponding seller. The search indicator is $x = 1$ if the buyer searches the second seller and $x = 0$ otherwise. Consumer surplus is the buyer's payoff. Total surplus includes both sellers and the platform:

$$\mathrm{CS} = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x, \tag{16}$$

$$\mathrm{TS} = \mathrm{CS} + \pi_1 + \pi_2 + \mathrm{PR} = \sum_{k=1}^{2} v_k \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \tag{17}$$

Prices and position payments are transfers. Search uses real resources, so total surplus depends on the product purchased and the search cost incurred. Search can increase surplus by revealing a valuable second product, but it can also incur a cost without changing the product ultimately purchased.

3 Experimental Design

3.1 Protocol

The experiment implements the three-stage market described above. Each market consists of two sellers and one buyer. Each product independently has a good match value of $V = 100$ experimental points with probability $q = 0.5$ and a bad match value of zero otherwise. I vary the cost of searching the second seller between sessions. The low-cost treatment has $s = 20$ and the high-cost treatment has $s = 40$. All other market rules are the same, and subjects participate in only one treatment.

Sellers bid for positions, learn their positions and the position payment, and then set prices. The buyer observes both prices and the top match before deciding whether to search the second seller. She can buy the second seller's product only after search and can return to the top seller for free. Bids and prices are restricted to integer values from 0 to 100. The continuous-price equilibrium provides the theoretical benchmark for this integer-grid implementation.

Roles are assigned at the beginning and remain fixed. In each round, the computer randomly and anonymously rematches subjects into groups of two sellers and one buyer and independently draws new match values. Subjects receive written instructions and answer three comprehension questions before completing two unpaid practice rounds and 30 incentivized rounds. At the end of each incentivized round, they receive feedback on the realized history of their market and their own payoff. Feedback concerns information revealed during play. The second match value is not disclosed when the buyer has not searched that product.

Every round begins with a balance of 200 experimental points, reset independently of earlier earnings. This balance covers possible losses from search and position payments. One of the 30 incentivized rounds is randomly selected for payment. The selected round's balance plus payoff is converted at 10 experimental points per dollar and added to a \$10 show-up fee. The welfare measures exclude the initial balance and show-up payment and measure surplus generated within the market. Appendix E reproduces the instructions.

3.2 Laboratory Implementation

I conducted six computerized sessions at the University of Virginia's Veconlab, with three sessions per treatment. There were 42 subjects in the low-cost treatment and 45 in the high-cost treatment. Five sessions had 15 subjects and one low-cost session had 12. The low-cost treatment included 28 sellers and 14 buyers, while the high-cost treatment included 30 sellers and 15 buyers. With 30 incentivized rounds per subject, the sample contains 420 low-cost and 450 high-cost market observations.

Each market in each round is one observation for the outcome comparisons, which treat these observations as independent. Search regressions use session-clustered standard errors.

3.3 Hypotheses

The hypotheses apply the continuous equilibrium to the experimental parameters. Numerical predictions average over equilibrium prices and match draws before their realization. Section 4.1 also compares observed outcomes with benchmarks conditional on the realized match draws. In both calculations, the buyer's decisions use only the information available at the time of choice. Appendix D gives the calculations.

Hypothesis 1 (Bidding and platform revenue). Raising the search cost from 20 to 40 raises equilibrium bids and platform revenue from 20 to 40.

The larger search disadvantage of the second seller increases the revenue advantage of the top position by 20. Competition for that position transfers the increase to the platform.

Hypothesis 2 (Prices). Top prices rise and second prices fall as search cost increases, with price distributions given by (9) and (10).

In the low-search-cost treatment, the top and second price supports are $[70, 100]$ and $[30, 60]$, with means of 81.59 and 49.93. In the high-search-cost treatment, the supports are $[90, 100]$ and $[10, 20]$, with means of 93.86 and 18.96. The top price distribution first-order stochastically dominates the second price distribution in each treatment.

Hypothesis 3 (Search). Search follows the match-specific cutoffs in Lemma 1 and becomes less frequent as search cost increases.

After a bad top match, the buyer searches when the second price is at most 60 in the low-search-cost treatment and at most 20 in the high-search-cost treatment. After a good top match, she searches when the price gap exceeds 40 and 80, respectively. At equilibrium prices, a buyer with a bad top match always searches. Search after a good top match occurs with probabilities 26.65% and 8.06%, so the overall search probabilities are 63.32% and 54.03%.

Hypothesis 4 (Revenues, profits, and welfare). Higher search cost raises the top seller's expected revenue and lowers the second seller's expected revenue, both sellers' expected profits, consumer surplus, and total surplus.

Expected top and second revenues change from 35 and 15 in the low-search-cost treatment to 45 and 5 in the high-search-cost treatment. Each seller's expected profit falls from 15 to 5, so combined seller profits fall from 30 to 10. Expected total surplus falls from 62.34 to 53.39 and expected consumer surplus falls from 12.34 to 3.39. These are unconditional predictions. The realized-draw benchmarks used in the empirical comparison can differ because the observed frequencies of match pairs need not equal their ex ante probabilities.

4 Experimental Results

I first compare welfare outcomes with equilibrium, then examine the bidding, pricing, and search decisions behind these outcomes. I also compare revenues and profits at the two positions to distinguish the return from occupying the top position from the return after paying for it. The analysis uses all 30 incentivized rounds. Unless stated otherwise, reported means pool the 420 low-cost and 450 high-cost markets, giving each market equal weight.

4.1 Welfare

For each market, consumer surplus is the value of the purchased product minus its price and any search cost. Platform revenue is the losing bid paid by the top seller. Combined seller profits equal the two sellers' sales revenues less that payment. Equation (17) gives the accounting identity: prices and position payments redistribute surplus, while allocation and search determine its total. A search cost is counted even when the buyer exits or returns to the top seller.

The equilibrium comparison uses the same match draws as the experiment. For each market, I calculate the expected equilibrium outcome conditional on its realized pair (v_1, v_2), integrating over the equilibrium price distributions, and then average these conditional predictions. Buyers in this calculation retain the model's information structure: they do not learn v_2 before deciding whether to search. Conditioning adjusts the comparison for the finite sample of match draws. It does not give the benchmark buyer additional information. In particular, the equilibrium platform payment remains s in every market. The observed-draw adjustment can make the two positions' sample-average equilibrium profits differ even though their ex ante equilibrium profits are equal.

Figure 5 and Table 1 report the welfare outcomes. Within each treatment, observed outcomes are paired with their market-specific equilibrium benchmarks. Across treatments, I use Welch tests. Each market in each round is treated as an independent observation for these tests. All tests are two-sided and unadjusted for multiple comparisons.

Total surplus. Observed total surplus falls from 58.62 in the low-search-cost treatment to 50.09 in the high-search-cost treatment, a decline of 8.53 ($p = 0.013$). The draw-conditional equilibrium benchmarks are 64.03 and 54.23, a decline of 9.80. Thus, the observed change has the predicted direction, while total surplus is below equilibrium in both treatments. The observed-minus-equilibrium gaps are -5.41 at the low cost and -4.15 at the high cost (paired $p < 0.001$ and $p = 0.004$).

The accounting in (17) clarifies these gaps. The mean allocated match value is 66.67 at the low cost and 60.22 at the high cost, compared with benchmarks of 76.90 and 75.11. Observed search expenditure is also lower: 8.05 versus 12.87, and 10.13 versus 20.88. Fewer

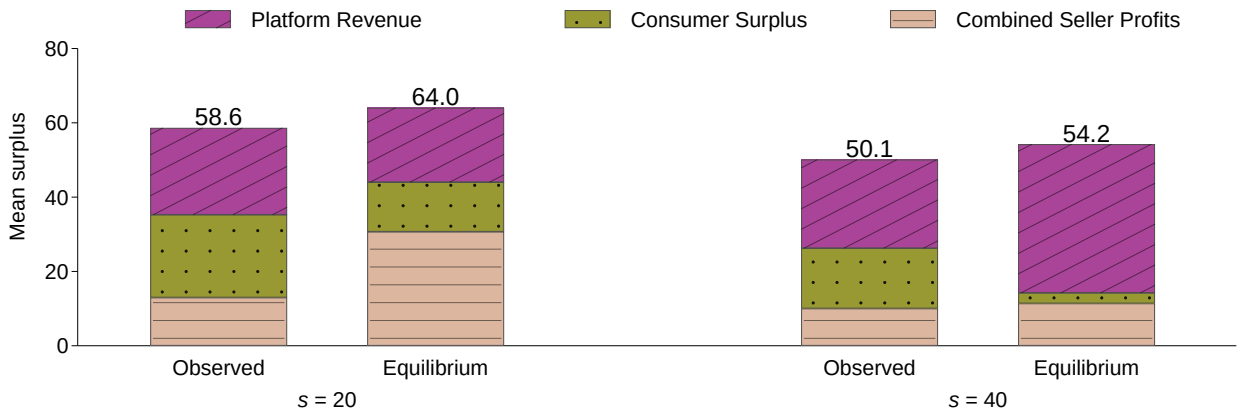


Figure 5: The level and distribution of surplus

Means in experimental points. Each bar is total surplus, divided into platform revenue, consumer surplus, and combined seller profits. Equilibrium outcomes condition on the observed match draws and integrate over equilibrium prices. Values above the bars report total surplus. Table 1 reports each component and comparison. The figure displays means without confidence intervals.

Table 1: Welfare outcomes and equilibrium benchmarks

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted
Total surplus	58.62***	64.03	50.09***	54.23	-8.53	-9.80
Platform revenue	23.35***	20.00	23.83***	40.00	0.48	20.00
Consumer surplus	22.29***	13.34	16.23***	2.84	-6.06	-10.50
Combined seller profits	12.98***	30.69	10.03	11.39	-2.95	-19.29

Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Theory conditions on each market's realized match values and averages over equilibrium pricing and buyer decisions. Seller profits subtract the payment for prominence.

resources are spent on search, but the reduction in realized allocation value is larger. This accounting shows how the two components of total surplus differ from their equilibrium values. It does not separately attribute these differences to prices and buyer choices.

Result 1. Total surplus is below the equilibrium benchmark in both treatments and decreases as search cost rises.

Platform revenue. The predicted increase of 20 in auction revenue does not appear in the pooled means. Platform revenue is 23.35 in the low-search-cost treatment and 23.83 in the high-search-cost treatment, an increase of 0.48 ($p = 0.718$). The 95% Welch interval for this increase is $[-2.11, 3.07]$. This estimate is small relative to the predicted increase of 20, although nonrejection of equal observed means does not establish equality. Revenue exceeds the low-cost equilibrium benchmark by 3.35 and falls below the high-cost benchmark by 16.17 (both market-level $p < 0.001$). The relevant auction statistic is the losing bid, which determines the payment, rather than the average submitted bid.

Result 2. Platform revenue changes little as search cost rises, despite the predicted increase of 20. It exceeds the equilibrium benchmark at the low search cost and falls below it at the high search cost.

Consumer surplus. Consumers receive more surplus than predicted in both treatments. Observed consumer surplus is 22.29 in the low-search-cost treatment, against a benchmark of 13.34, and 16.23 in the high-search-cost treatment, against 2.84 (both paired $p < 0.001$). It falls by 6.06 across treatments ($p = 0.002$), compared with a predicted decline of 10.50.

The proportional comparison is sharper: observed consumer surplus falls by 27.2%, whereas its draw-conditional equilibrium benchmark falls by 78.7%. These percentages are changes in treatment means, not averages of individual percentage changes. A delta-method comparison, retaining the covariance between observed and equilibrium surplus within each treatment, gives $p < 0.001$ for the 51.5-percentage-point difference. The higher level of consumer surplus is compatible with lower total surplus: consumers make lower payments and incur less search expenditure, while some valuable purchases fail to occur.

Result 3. Consumer surplus exceeds the equilibrium benchmark in both treatments. It decreases with search cost, but the proportional decline is smaller than predicted.

Combined seller profits. The two sellers together earn 12.98 after the position payment in the low-search-cost treatment, well below the equilibrium benchmark of 30.69 (paired $p < 0.001$). In the high-search-cost treatment, their mean profit is 10.03 against a benchmark of 11.39. The shortfall of 1.36 is comparatively small and is not statistically detected ($p = 0.358$). The across-treatment difference is also imprecisely estimated ($p = 0.233$). Sellers

Table 2: Bids and posted prices

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted
Top price	61.83***	81.59	66.07***	93.86	4.24	12.27
Second price	48.90	49.93	39.11***	18.96	-9.79	-30.97
Pooled bid	34.03***	20.00	38.79	40.00	4.76	20.00
Winning bid	44.71***	20.00	53.76***	40.00	9.05	20.00
Losing bid	23.35***	20.00	23.83***	40.00	0.48	20.00

Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Theory prices are analytical equilibrium means and theory bids equal the search cost. The pooled bid is the mean of the two submitted bids in a market and contributes one observation. The losing bid equals the top seller's payment to the platform.

retain 22.1% and 20.0% of observed total surplus, respectively. These shares are ratios of treatment means, not averages of market-specific ratios.

The proximity to equilibrium profit at the high search cost does not imply that seller behavior or revenues are close to equilibrium. As Table 4 shows, combined sales revenue is 36.33 rather than 50.69 at the low cost and 33.86 rather than 51.39 at the high cost. Revenue at each position is below its draw-conditional benchmark at both costs (all paired market-level $p < 0.001$). The profit identity explains the different profit gaps. In the low-search-cost treatment, a revenue shortfall of 14.36 and an auction payment 3.35 above equilibrium together produce the profit shortfall of 17.71. In the high-search-cost treatment, the revenue shortfall of 17.53 is largely offset by an auction payment 16.17 below equilibrium, leaving a profit shortfall of 1.36. This is an exact accounting relation, not a decomposition into independently identified behavioral effects.

Result 4. The sellers jointly retain about one fifth of total surplus after paying for prominence. Their combined profit is below equilibrium at the low search cost and is much closer to the benchmark at the high search cost.

4.2 Bidding, Pricing, and Search

Bidding. Figure 6 and Table 2 distinguish pooled, winning, and losing bids. The mean submitted bid rises from 34.03 to 38.79, an increase of 4.76. At the market level, this increase differs from zero and is smaller than the predicted increase of 20 (both $p < 0.001$). For this test, the pooled-bid observation is the average of the two bids within a market, so the two bids are not treated as separate observations. The pooled mean is above the equilibrium bid of 20 in the low-cost treatment ($p < 0.001$). Its difference from 40 in the high-cost treatment is not detected ($p = 0.221$).

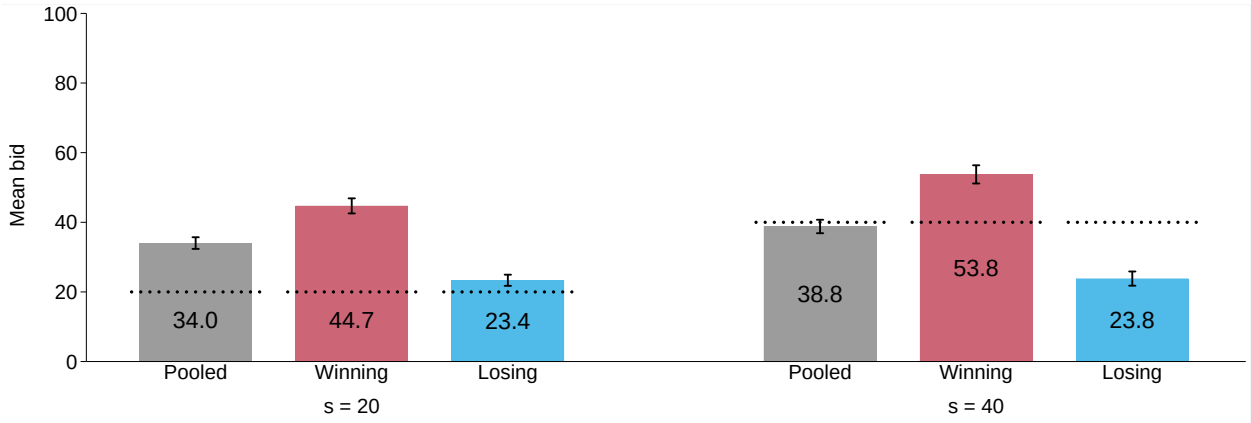


Figure 6: Pooled, winning, and losing bids

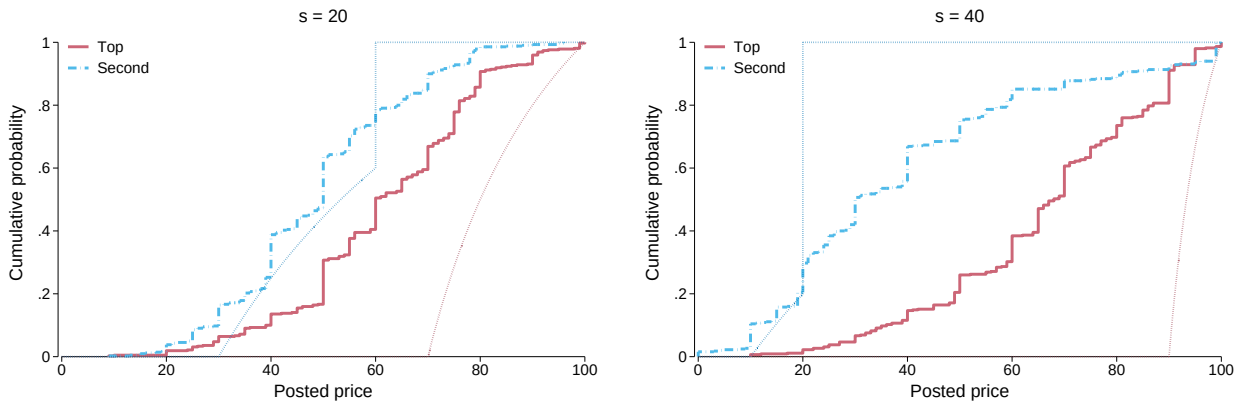
Means over all 30 rounds. The pooled bid is the mean of both bids in a market. The winning bid determines the top position, while the losing bid determines the payment to the platform. Error bars show 95% confidence intervals treating markets as independent. Horizontal reference lines show the equilibrium bids of 20 and 40.

Winning bids rise from 44.71 to 53.76, an increase of 9.05 rather than the predicted 20. Losing bids remain near 24. Since the top seller pays the losing bid, the increase in winning bids does not itself increase platform revenue. The high-cost pooled mean also conceals substantial differences between winning and losing bids: a mean near 40 need not imply that individual bids concentrate near the equilibrium bid.

Result 5. Mean submitted bids rise by 4.76 as search cost increases, well short of the predicted increase of 20. Winning bids rise, while losing bids change little.

Pricing. Figure 7 compares observed and equilibrium price distributions. Top sellers post substantially lower prices than predicted: their means are 61.83 and 66.07, compared with equilibrium means of 81.59 and 93.86 (one-sample market-level $p < 0.001$ at both costs). The second seller's mean is 48.90 against 49.93 at the low cost ($p = 0.196$), but 39.11 against 18.96 at the high cost ($p < 0.001$). These tests compare observed prices with the fixed analytical equilibrium means. The posted-price distributions also extend well beyond the equilibrium supports. Top-price standard deviations are 17.48 and 20.27, against predicted values of 8.39 and 2.80. Second-price standard deviations are 16.30 and 25.36, against 10.57 and 2.44. These dispersion comparisons are descriptive.

Top prices exceed second prices in 66.7% of low-cost markets and 81.6% of high-cost markets. The corresponding mean gaps are 12.93 and 26.96, below the predicted 31.65 and 74.90. These paired comparisons establish neither a uniform price ordering nor first-order stochastic dominance: the empirical cumulative distributions cross. More importantly, an average price gap does not determine search after a bad top match. In that case, the relevant comparison is the second price with $100 - 2s$, independently of the top price.



(a) Low search cost

(b) High search cost

Figure 7: Observed and equilibrium price distributions

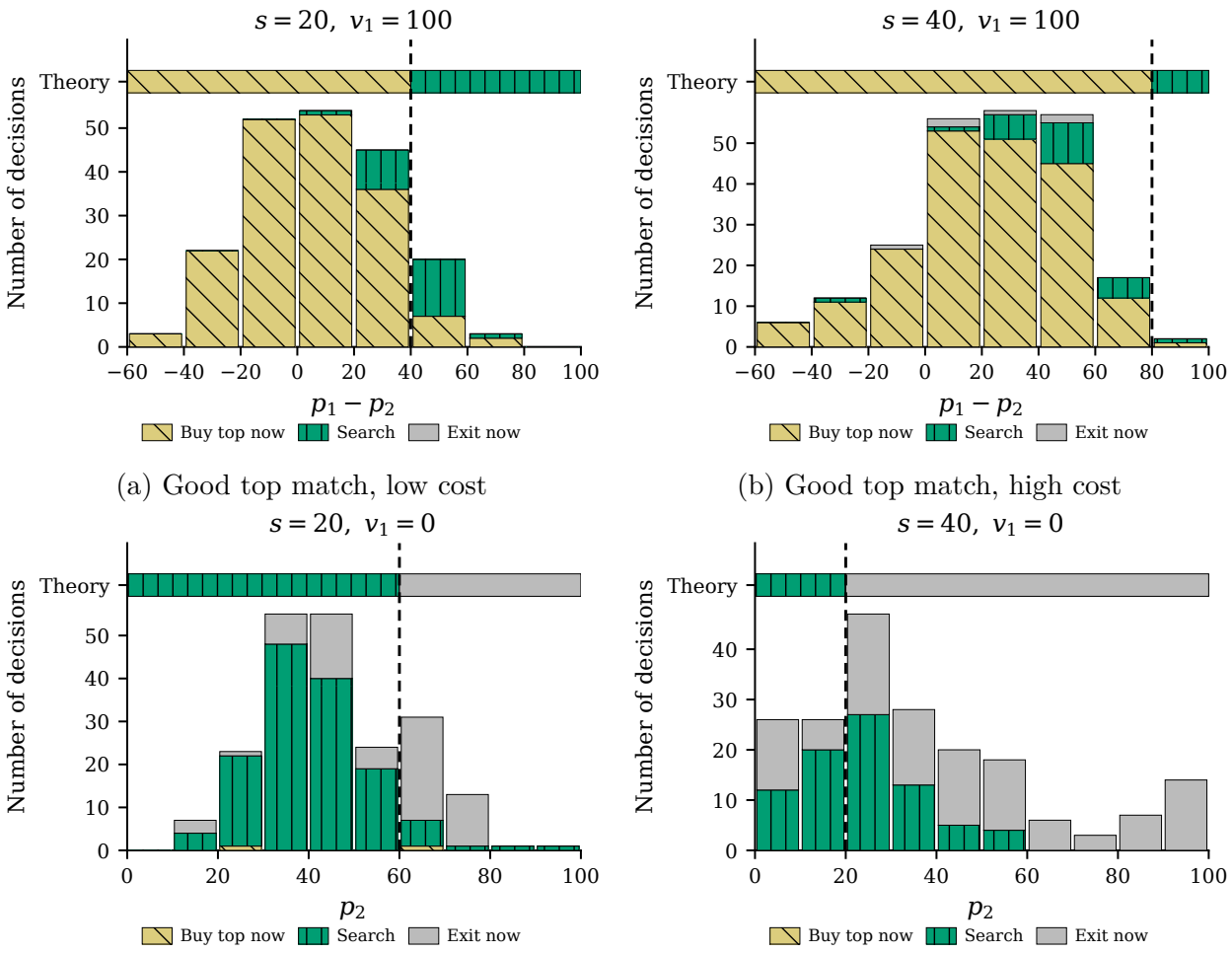
All 30 rounds. Red and blue distinguish the top and second sellers. The solid red and dash-dot blue curves show the observed CDFs of top and second prices, respectively. Dotted curves show the corresponding equilibrium CDFs. These distributions concern posted prices, whether or not a sale occurs.

The second seller exceeds that bad-match search cutoff in 21.4% of all low-cost markets and 70.2% of all high-cost markets. Among buyers who actually receive a bad top match, the second price lies strictly above the cutoff in 46 of 216 low-cost observations and 143 of 216 high-cost observations. At those prices, stopping is optimal. The implications for sales and buyer surplus depend on the search choices discussed next.

Result 6. Prices at both positions are more dispersed than predicted, with observed prices extending beyond the equilibrium supports. Top sellers charge higher prices on average than second sellers, but lower prices than predicted. Second sellers charge more than predicted at the high search cost.

Search decisions. I evaluate search choices at the prices subjects actually observe. With a good top match, search is strictly optimal when $p_1 - p_2 > 2s$. With a bad top match, it is strictly optimal when $p_2 < 100 - 2s$. Figure 8 displays decisions around these cutoffs. Table 3 excludes exact search ties when calculating compliance and reports their counts separately. This avoids classifying an indifferent buyer's search choice as a strict error. The equilibrium tie convention remains relevant to the theoretical benchmark, but the empirical accuracy measure does not require subjects to follow that convention.

After a good top match, 90.5% of low-cost and 89.7% of high-cost decisions satisfy the strict search rule. These high rates partly reflect the prices buyers face. In the high-cost treatment, 231 of the 233 strict cases call for stopping, leaving only two cases in which search is strictly preferable. Overall compliance therefore provides limited evidence about behavior on both sides of that cutoff. After a bad top match, compliance is lower: 80.5% in the low-search-cost treatment and 64.6% in the high-search-cost treatment.



(a) Good top match, low cost

(b) Good top match, high cost

(c) Bad top match, low cost

(d) Bad top match, high cost

Figure 8: Search decisions at observed prices

All 30 rounds. Bar heights show the frequency of observed prices or price differences. Segments distinguish buying from the top seller, searching the second seller, and exit at the first decision. The price difference governs search after a good top match. The second price governs it after a bad top match. The strip marked Theory indicates the optimal action in each price region. Cutoff ties are shown in the figures but excluded from the strict-compliance rates in Table 3.

Table 3: Search decisions relative to the optimal cutoff

Top match	Search cost	Strict cases	Compliant	Under-search	Over-search	Ties
Good	$s = 20$ (Low)	199	90.5%	9	10	5
Good	$s = 40$ (High)	233	89.7%	1	23	1
Bad	$s = 20$ (Low)	210	80.5%	32	9	6
Bad	$s = 40$ (High)	195	64.6%	20	49	21

Compliance classifies the search decision at observed prices. Exact search ties are excluded from the strict cases and reported separately. Under-search means stopping when search has strictly positive net expected value. Over-search means searching when stopping is strictly preferable. Counts of errors are not conditional error rates. Purchase optimality is a separate decision.

Errors occur in both directions. Low-cost buyers with a bad top match fail to search in 32 of the 164 cases in which search is strictly optimal, and search in 9 of the 46 cases in which stopping is optimal. At the high cost, the corresponding counts are 20 of 52 and 49 of 143. The conditional under-search rates are therefore 19.5% and 38.5%, and the over-search rates are 19.6% and 34.3%. Under-search can prevent a valuable second purchase. Over-search incurs an search cost when it is not justified by the buyer's expected private gain. Search accuracy is distinct from purchase accuracy after the available products have been revealed.

Purchase choices are much closer to the payoff-maximizing choice within the buyer's chosen information set: there are four purchase errors among 420 low-cost markets and six among 450 high-cost markets. The principal departures therefore concern how sellers bid and price and whether buyers search the second product, rather than which searched product buyers choose.

Result 7. Most search decisions satisfy the optimal cutoff after a good top match. Compliance is lower after a bad top match, particularly at the high search cost, with both under-search and over-search.

Pricing and search jointly explain how consumers can receive more surplus while sellers earn less revenue. Lower top prices raise the surplus from buying the top product. After a good top match, they also reduce the price gap and can make stopping optimal, saving the buyer a further search cost. The same lower prices reduce the top seller's revenue per sale.

After a bad top match, the second price determines whether further search is worthwhile. Prices above the cutoff discourage search even when the second product would turn out to be a good match. This is an optimal response to the posted price, rather than a buyer mistake. Under-search at prices below the cutoff can prevent additional valuable purchases. Over-search, in contrast, incurs a cost when the expected private gain is too small. These different responses must be distinguished when interpreting the lower sales revenues and total surplus.

The payment for prominence then determines how much of the sellers' revenue remains as profit. At the low search cost, lower revenue and an above-equilibrium payment both reduce seller profits. At the high search cost, the payment shortfall offsets much of the revenue shortfall. Similar profits can therefore conceal very different pricing and bidding behavior.

4.3 Revenues and profits by position

Combined profits conceal how revenues and payments differ between positions. Table 4 separates the two sellers' revenues and profits. The top seller earns more revenue on average in both treatments, but it is also the only seller that pays the platform.

The observed revenue advantage of the top position is 19.01 at the low search cost and 27.91 at the high search cost. The corresponding mean payments are 23.35 and 23.83. Thus,

Table 4: Revenues and profits by position

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted
Top revenue, Π_1	27.67***	34.17	30.89***	46.73	3.22	12.57
Second revenue, Π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86
Combined revenue	36.33***	50.69	33.86***	51.39	-2.47	0.71
Revenue advantage, $\Pi_1 - \Pi_2$	19.01	17.64	27.91***	42.07	8.91	24.43
Platform revenue	23.35***	20.00	23.83***	40.00	0.48	20.00
Top profit, π_1	4.32***	14.17	7.06	6.73	2.74	-7.43
Second profit, π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86
Profit difference, $\pi_1 - \pi_2$	-4.34	-2.36	4.09	2.07	8.43	4.43

Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Theory conditions on realized match values. Revenue excludes the prominence payment and top profit subtracts it. Position differences describe auction-assigned positions and do not identify a causal effect of moving a seller between positions.

at the low cost, the payment exceeds the observed revenue advantage. Average top profit is 4.32, compared with 8.66 for the second seller. At the high cost, top profit is 7.06 and second profit is 2.97. Average revenue is higher in the top position, but average profit need not be higher after paying for it.

These comparisons satisfy the market-by-market identity

$$\pi_1 - \pi_2 = (\Pi_1 - \Pi_2) - b^{(2)}. \tag{18}$$

The distinction between a bid and a payment is also important. Mean submitted bids of 34.03 and 38.79 exceed the observed revenue advantages of 19.01 and 27.91. The second-price auction does not charge the winning seller its own bid, so this comparison alone does not measure a realized loss from bidding.

Nor is the observed revenue advantage a direct estimate of any seller's willingness to pay. Sellers are selected into positions by their bids, and they bid before observing the subsequent prices and match values. Evaluating an individual bid requires that seller's expected continuation revenues at the time of bidding. The position-specific averages describe the returns realized in these markets. They do not establish what the same seller would have earned in the other position.

5 Search regressions

The cutoff comparisons describe whether buyers search when search is optimal at the observed prices. I now examine how the probability of search varies with each price, conditional on

the top match. This separates the response to the second price after a bad top match from the response to the price gap after a good top match.

Let $G = 1$ if the top product is a good match and $G = 0$ if it is a bad match. Let r denote the round. For each search cost s, I estimate a separate logit model using all 30 incentivized rounds:

$$\Pr(x = 1 \mid G, p_1, p_2, r, s) = \Lambda(\eta_s),$$
$$\eta_s = \alpha_s + \gamma_s G + \beta_{1s} p_1 + \beta_{2s} p_2 + \delta_{1s} G p_1 + \delta_{2s} G p_2 + \rho_s r. \tag{19}$$

Here Λ is the logistic distribution function. The interactions $G \times p_1$ and $G \times p_2$ allow the two price slopes to differ by top match. After a bad match, the price coefficients in the logit index are β_{1s} and β_{2s}. After a good match, they are $\beta_{1s} + \delta_{1s}$ and $\beta_{2s} + \delta_{2s}$. Every coefficient can differ by treatment. The round trend is common to the two match states within a treatment. The model uses information available when the buyer chooses whether to search and excludes the hidden second match and later purchase outcome. The estimates describe conditional price associations rather than causal price effects.

The theory suggests two restrictions on the index:

$$\begin{aligned} H_B: &\quad \beta_{1s} = 0, \\ H_G: &\quad \beta_{1s} + \delta_{1s} + \beta_{2s} + \delta_{2s} = 0. \end{aligned} \tag{20}$$

After a bad top match, the top price should not affect search. After a good top match, dependence on the price gap requires the two price coefficients to have equal magnitudes and opposite signs. These restrictions concern the specified index and do not impose the theory's deterministic cutoffs.

Table 5 reports the logit coefficients, with standard errors clustered by session.[4] The sample contains 420 low-cost and 450 high-cost decisions.

To express the price associations in probability units, I calculate average marginal effects from the fitted model separately for each top match. I evaluate each match state at every observed price and round vector within its treatment, setting G to zero or one, and average the resulting price derivatives. The effects below are percentage points per 10 experimental points of price. They are scaled local derivatives rather than exact probability changes from a finite increase of 10 experimental points.

After a bad top match, a higher second price is associated with substantially less search. The average derivatives are -12.57 and -9.06 percentage points per 10 experimental points of price at the low and high search costs. The top-price derivatives are much smaller, at

[4]There are only three sessions in each treatment, so the nominal session-clustered Wald p-values are unreliable as calibrated measures of sampling uncertainty and are sensitive to the variance estimate.

Table 5: Search decisions: logit regression coefficients

	$s = 20$ (Low)	$s = 40$ (High)
Top price, p_1	0.0113	0.0105
	(0.0103)	(0.0089)
Second price, p_2	-0.0716	-0.0458
	(0.0042)	(0.0067)
Good top match, G	-5.6709	-3.8153
	(0.4016)	(0.6062)
$G \times p_1$	0.0689	0.0248
	(0.0047)	(0.0080)
$G \times p_2$	-0.0616	0.0054
	(0.0077)	(0.0092)
Round	-0.0116	-0.0281
	(0.0103)	(0.0350)
Constant	3.7778	0.9655
	(1.3355)	(0.5842)
Observations	420	450
Sessions	3	3

The dependent variable equals one if the buyer searches the second seller's product. Each column is a separate treatment regression using all 30 incentivized rounds. Parentheses contain standard errors clustered by session. Prices enter in experimental points. The entries are logit coefficients, not changes in search probability. G equals one for a good top match and zero for a bad top match.

1.99 and 2.08 percentage points. The direct tests of H_B do not reject under session-clustered inference, with nominal p-values of 0.2696 and 0.2390.

After a good top match, a higher top price is associated with more search and a higher second price with less search. The derivatives are 5.56 and -9.23 at the low cost, compared with 3.09 and -3.54 at the high cost. Raising both prices equally while keeping their gap fixed therefore has an implied average derivative of -3.67 percentage points per 10 experimental points of price at the low cost and -0.45 at the high cost. The latter point estimate is economically much closer to the gap-only restriction.

The session-clustered logit tests reject H_G in both treatments, with nominal p-values of 0.0015 and 5.63×10^{-6}. The price responses have the signs predicted by the search incentives, but these tests suggest that search does not depend solely on the price gap after a good match. With only three session clusters per treatment, the rejection should be interpreted cautiously.

6 Conclusion

This paper studies a market in which sellers compete for prominence before setting prices and consumers decide how far to search. The model links the value of the prominent position

to the pricing and search decisions that follow its allocation. A laboratory experiment varies the cost of searching the second seller's product and observes how surplus is divided among consumers, sellers, and the platform.

The welfare results have three main features. Platform auction revenue changes little across search costs, reflecting the weak response of losing bids. Consumer surplus exceeds its equilibrium benchmark at both costs and falls proportionally less than predicted as the cost increases. Sellers jointly retain only about one fifth of total surplus after paying for prominence. Lower top prices benefit buyers who purchase from the top seller but reduce seller revenue per sale. Second prices above the search cutoff can discourage further search after a bad initial match and lead to lost sales. Combined sales revenue is below equilibrium in both treatments, and position payments take a substantial share of that revenue. At the high search cost, payments below equilibrium offset much of the revenue shortfall, explaining why profits are closer to the benchmark there.

The experiment separates the search-cost advantage of prominence from information that rankings may convey in the field. It also shows why bids and product-market outcomes should be studied together. Auction payments depend on sellers' valuations of prominence, while those valuations depend on anticipated pricing and consumer search. An auction model that takes position values as given can describe payment incentives, but cannot alone explain how a change in consumer search friction changes those values. Conversely, a model that assigns prominence without payment misses the transfer of seller revenue to the platform.

These findings suggest several directions for further work. One concerns platforms' incentives to change search costs. In the experiment, a lower cost improves consumer and total surplus without a statistically detectable decline in auction revenue. This comparison does not establish an optimal search cost for a platform: investment in search technology may be costly, and transaction commissions, participation, and other revenue sources may also respond. An endogenous platform decision would need to incorporate these margins rather than equate platform profit with auction revenue alone.

A second question is how prominence should be allocated. Comparing the present auction with random allocation or a first-price auction would show how outcomes depend on the allocation and payment mechanisms. With more sellers and several degrees of prominence, the allocation could also affect which products enter consumers' consideration sets. Seller differences in quality, costs, or match distributions would then matter for whether willingness to pay selects the socially preferred search order. The two identical sellers in this paper isolate the value of a search-cost advantage. They do not settle that broader allocation question.

A third question concerns the product information that sellers can display to consumers without further search. Here, prices are visible but match values must be learned through search. Other information structures may change how sellers direct attention and how buyers choose which products to search. Studying price visibility alongside product characteristics,

reviews, and platform disclosure rules would help connect competition for prominence to the design of the search environment itself.

References

Abrams, E., M. Sefton, and A. Yavas (2000): "An Experimental Comparison of Two Search Models," *Economic Theory*, 16, 735–749.

Anderson, S. P. and R. Renault (1999): "Pricing, Product Diversity, and Search Costs: A Bertrand–Chamberlin–Diamond Model," *The RAND Journal of Economics*, 30, 719–735.

——— (2026): "Search Direction: Position Externalities and Position Auction Bias," *The Review of Economic Studies*, 93, 763–797.

Armstrong, M. (2017): "Ordered Consumer Search," *Journal of the European Economic Association*, 15, 989–1024.

Armstrong, M., J. Vickers, and J. Zhou (2009): "Prominence and Consumer Search," *The RAND Journal of Economics*, 40, 209–233.

Armstrong, M. and J. Zhou (2011): "Paying for Prominence," *The Economic Journal*, 121, F368–F395.

Athey, S. and G. Ellison (2011): "Position Auctions with Consumer Search," *The Quarterly Journal of Economics*, 126, 1213–1270.

Brown, A. L., A. Viriyavipart, and X. Wang (2018): "Search Deterrence in Experimental Consumer Goods Markets," *European Economic Review*, 104, 167–184.

Cason, T. N. and D. Friedman (2003): "Buyer Search and Price Dispersion: A Laboratory Study," *Journal of Economic Theory*, 112, 232–260.

Chen, Y. and C. He (2011): "Paid Placement: Advertising and Search on the Internet," *The Economic Journal*, 121, F309–F328.

Choi, M., A. Y. Dai, and K. Kim (2018): "Consumer Search and Price Competition," *Econometrica*, 86, 1257–1281.

Davis, D. D. and C. A. Holt (1996): "Consumer Search Costs and Market Performance," *Economic Inquiry*, 34, 133–151.

Diamond, P. A. (1971): "A Model of Price Adjustment," *Journal of Economic Theory*, 3, 156–168.

Ding, Y. and T. Zhang (2018): "Price-Directed Consumer Search," *International Journal of Industrial Organization*, 58, 106–135.

Edelman, B., M. Ostrovsky, and M. Schwarz (2007): "Internet Advertising and the Generalized Second-Price Auction: Selling Billions of Dollars Worth of Keywords," *American Economic Review*, 97, 242–259.

Obradovits, M. and P. Plaickner (2023): "Price-Directed Search, Product Differentiation and Competition," *Review of Industrial Organization*, 63, 317–348.

Ursu, R. M. (2018): "The Power of Rankings: Quantifying the Effect of Rankings on Online Consumer Search and Purchase Decisions," *Marketing Science*, 37, 530–552.

Varian, H. R. (2007): "Position Auctions," *International Journal of Industrial Organization*, 25, 1163–1178.

Wolinsky, A. (1986): "True Monopolistic Competition as a Result of Imperfect Information," *The Quarterly Journal of Economics*, 101, 493–511.

Xu, L., J. Chen, and A. B. Whinston (2011): "Price Competition and Endogenous Valuation in Search Advertising," *Journal of Marketing Research*, 48, 566–586.

Zhou, J. (2011): "Ordered Search in Differentiated Markets," *International Journal of Industrial Organization*, 29, 253–262.

Appendix A Derivation of buyer search and demand

A.1 Choices at indifference

The buyer behavior in Lemma 1 uses the following choices when several actions are optimal.

A1. With $v_1 = V$, the buyer stops when indifferent between searching the second seller and stopping.

A2. With $v_1 = 0$, the buyer searches the second seller when indifferent between searching and stopping.

A3. Among available choices yielding the same highest purchase surplus, the buyer prefers purchase from the top seller to purchase from the second seller, and purchase from the second seller to exit.

These rules select optimal buyer actions at equality and support the pricing equilibrium in Theorem 1. Figure 9 gives the complete buyer decision tree conditional on the observed prices and top match.

A.2 Proof of Lemma 1

At any $p_1, p_2 \geq 0$, stopping gives $\max\{0, v_1 - p_1\}$. The expected gain from search before paying its cost is

$$q \max\{0, v_1 - p_1, V - p_2\} + (1 - q) \max\{0, v_1 - p_1, -p_2\} - \max\{0, v_1 - p_1\}. \tag{21}$$

The buyer searches when this gain exceeds s and stops when it is smaller. A1 and A2 resolve equality. Once search is complete, the search cost is sunk, and A3 selects a surplus-maximizing terminal action.

For $0 \leq p_1, p_2 \leq V$ and $v_1 = 0$, the gain is $q(V - p_2)$, so A2 gives search exactly when $p_2 \leq V - \frac{s}{q}$. For $v_1 = V$, the gain is $q \max\{p_1 - p_2, 0\}$, so A1 gives search exactly when $p_1 - p_2 > \frac{s}{q}$. These prove the two cutoff rules. A price $p_1 > V$ yields no sales. If $p_2 > V - \frac{s}{q}$, the gain from search is at most $q \max\{V - p_2, 0\} < s$, so seller 2 is never searched. □

A.3 Derivation of seller demand

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the following paths give the demand functions in (6).

(v_1, v_2)	$p_1 - p_2 > \frac{s}{q}$	$p_1 - p_2 \leq \frac{s}{q}$
$(0, 0)$	Search, then exit	Search, then exit
$(0, V)$	Search, buy 2	Search, buy 2
$(V, 0)$	Search, return to 1	Buy 1 immediately
(V, V)	Search, buy 2	Buy 1 immediately

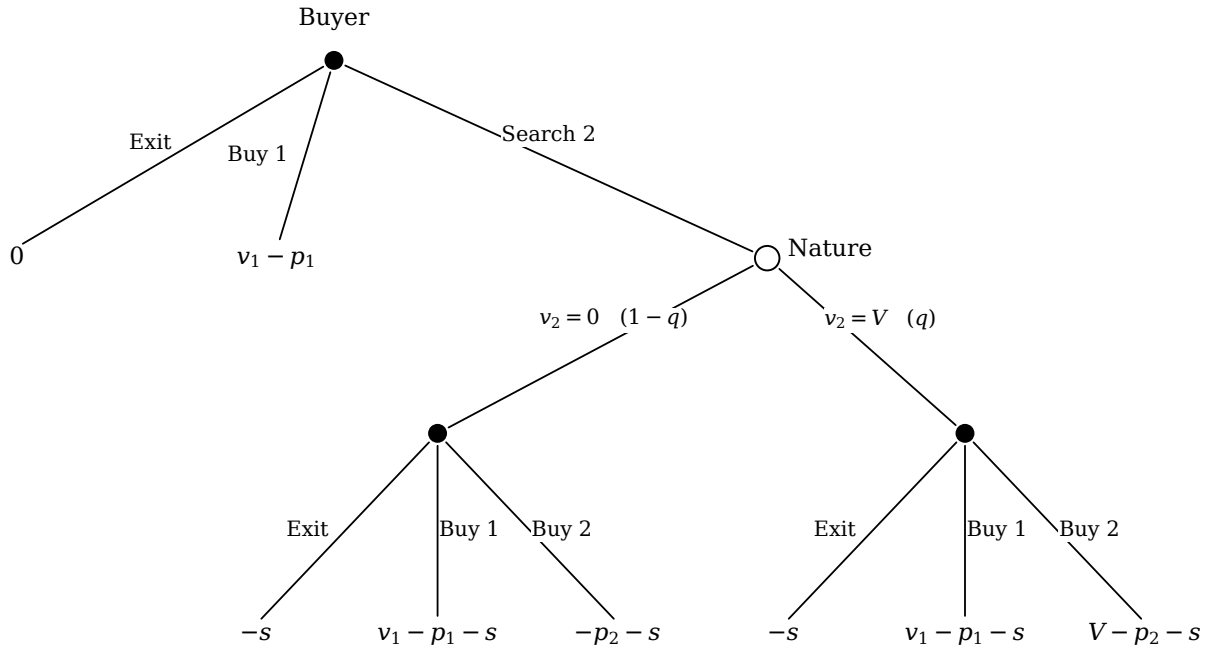


Figure 9: The buyer's complete decision tree

The buyer observes both prices and the top match before deciding whether to search the second seller. Search reveals the second match and costs s.

Appendix B Pricing equilibrium and price moments

B.1 Proof of Proposition 1

Each seller can guarantee strictly positive revenue. Seller 1 can charge any sufficiently small $0 < p_1 < \frac{s}{q}$ and sell whenever $v_1 = V$. Seller 2 can charge any $0 < p_2 < V - \frac{s}{q}$ and sell whenever $(v_1, v_2) = (0, V)$. Thus a pure equilibrium cannot involve a zero price, $p_1 > V$, or $p_2 > V - \frac{s}{q}$.

Consider the remaining domain $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$. If $p_1 - p_2 > \frac{s}{q}$, seller 2 can raise its price slightly while retaining demand q. If $p_1 - p_2 < \frac{s}{q}$, seller 1 can raise its price slightly while retaining demand q. At equality, seller 2 can choose a price strictly between $(1-q)p_2$ and p_2, increasing demand from $q(1-q)$ to q and strictly increasing revenue. Every candidate price pair therefore admits a profitable deviation. □

B.2 Proof of Theorem 1

The price bounds in Theorem 1 are

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \overline{p}_1 &= V, \\ \underline{p}_2 &= (1-q)\left(V - \frac{s}{q}\right), & \overline{p}_2 &= V - \frac{s}{q}. \end{aligned}$$

Both proposed CDFs are increasing on their continuous parts. The first rises from zero to one. The second starts at zero and approaches $1 - \frac{s}{qV}$ at its upper bound, with the remaining mass placed at that bound. Since $0 < s < qV$, both are valid distributions with strictly positive lower bounds.

Seller 1. For $0 < p_1 \leq V$, the expected revenue against F_2 is

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - q \Pr\left(p_2 < p_1 - \frac{s}{q}\right)\right]. \tag{22}$$

For $\underline{p}_1 \leq p_1 < V$, substitution of F_2 gives

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - \left(1 - \frac{(1-q)V + s}{p_1}\right)\right] = q[(1-q)V + s]. \tag{23}$$

At $p_1 = V$, a good-top-match buyer stops when $p_2 = V - \frac{s}{q}$ and searches at every lower second price. Including the mass at seller 2's upper bound therefore gives

$$V\left[q\frac{s}{qV} + q(1-q)\left(1 - \frac{s}{qV}\right)\right] = q[(1-q)V + s]. \tag{24}$$

A3 preserves seller 1's sales at zero purchase surplus. For $0 < p_1 < \underline{p}_1$, good-top-match buyers never search, so revenue is $qp_1 < q[(1-q)V + s]$. At $p_1 = 0$ or $p_1 > V$, revenue is zero. Hence every price in seller 1's support is a best response against all nonnegative pure-price deviations.

Seller 2. For $0 < p_2 \leq V - \frac{s}{q}$, bad-top-match buyers search. Good-top-match buyers search if $p_1 > p_2 + \frac{s}{q}$. Thus

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - qF_1\left(p_2 + \frac{s}{q}\right)\right]. \tag{25}$$

For $\underline{p}_2 \leq p_2 \leq \overline{p}_2$, substituting F_1 yields

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - \left(1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_2}\right)\right] = q(1-q)\left(V - \frac{s}{q}\right). \tag{26}$$

This calculation includes the upper bound: A2 ensures that bad-top-match buyers still search there. For $0 < p_2 < \underline{p}_2$, both top-match types search at every price in seller 1's support, giving revenue $qp_2 < q(1-q)(V - \frac{s}{q})$. A zero price gives zero revenue, and a price above $V - \frac{s}{q}$ attracts no search. Hence every price in seller 2's support is a best response against all nonnegative pure-price deviations.

The payoff from any mixed-price deviation averages the corresponding pure-price payoffs and cannot exceed these bounds. Both optimality conditions in Definition 1 therefore hold.□

B.3 Expected prices

$$\begin{aligned} \mathbb{E}[p_1] &= \frac{s}{q} + \frac{(1-q)\left(V - \frac{s}{q}\right)}{q} \log\left(\frac{1}{1-q}\right), \\ \mathbb{E}[p_2] &= \frac{(1-q)V + s}{q} \log\left(\frac{V}{(1-q)V + s}\right). \end{aligned} \tag{27}$$

Appendix C Bidding for prominence

C.1 Proof of Theorem 2

The two indifference calculations in (23) and (26) give the continuation revenues in (13). Their difference is

$$\begin{aligned}\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] &= q[(1-q)V + s] - q(1-q)\left(V - \frac{s}{q}\right) \\ &= qs + (1-q)s = s. \end{aligned} \tag{28}$$

Against an opponent bid b, winning rather than losing changes expected net profit by $s - b$. Bidding s wins when this difference is positive, loses when it is negative, and leaves the seller indifferent at equality. It is therefore weakly dominant conditional on the specified pricing continuation. With both sellers bidding s, the position payment is s and

$$\mathbb{E}[\pi_1] = q[(1-q)V + s] - s = (1-q)(qV - s) = \mathbb{E}[\pi_2].$$

The buyer is optimal after every price pair, and Theorem 1 supplies a pricing equilibrium after every auction outcome. These strategies and the symmetric bids therefore form a subgame-perfect equilibrium. □

Appendix D Calculation of revenues and welfare

D.1 Observed outcomes

For each market, the purchase indicator a and search indicator x determine consumer surplus and total surplus through (16) and (17). Seller revenue is its posted price if it makes the sale and zero otherwise. Platform revenue is the losing bid. Top profit subtracts that payment from top revenue, while second profit equals second revenue. The accounting identity is checked separately for every market before treatment means are calculated. Initial balances and show-up fees are excluded from all these measures.

To test an observed outcome against its conditional equilibrium benchmark, I first calculate their difference within each market. The within-treatment comparison is a one-sample test of whether the mean difference is zero. To test whether the treatment change matches the predicted change, I compare these differences across treatments with a Welch test. This preserves the pairing between an observed outcome and the benchmark based on the same match draws.

D.2 Benchmarks conditional on realized matches

The expectations below integrate over independent equilibrium prices, conditional on the indicated match draws. The price means are given in (27). For the calculation below, the lower price bounds are

$$\underline{p}_1 = (1-q)V + s, \qquad \underline{p}_2 = (1-q)\left(V - \frac{s}{q}\right).$$

After a good top match, search occurs when $p_1 - p_2 > \frac{s}{q}$. To calculate its probability, I integrate over the independent equilibrium price distributions in (9) and (10):

$$\begin{aligned}\Pr\left(p_1 - p_2 > \frac{s}{q}\right) &= \int_{\underline{p}_1}^{V} F_2\left(p_1 - \frac{s}{q}\right) dF_1(p_1) \\ &= \frac{\underline{p}_2}{q^2}\int_{\underline{p}_1}^{V} \frac{1 - \frac{\underline{p}_1}{p_1}}{\left(p_1 - \frac{s}{q}\right)^2}\, dp_1.\end{aligned}$$

The atom at the second seller's upper price bound contributes no search probability, since search at $p_2 = V - \frac{s}{q}$ would require $p_1 > V$. The top seller's distribution is atomless, so the value of the integrand at $p_1 = V$ does not affect the integral.

The logarithm comes from the reciprocal terms in the decomposition

$$\frac{1-\frac{\underline{p}_1}{p_1}}{\left(p_1-\frac{s}{q}\right)^2}=\frac{q^2\underline{p}_1}{s^2}\left(\frac{1}{p_1-\frac{s}{q}}-\frac{1}{p_1}\right)+\frac{1-\frac{q\underline{p}_1}{s}}{\left(p_1-\frac{s}{q}\right)^2}.$$

Integrating gives

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\left[\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{p_1-\frac{s}{q}}{p_1}\right)-\frac{\underline{p}_2}{q^2}\frac{1-\frac{q\underline{p}_1}{s}}{p_1-\frac{s}{q}}\right]_{p_1=\underline{p}_1}^{p_1=V}.$$

Using $\underline{p}_1-\frac{s}{q}=\underline{p}_2=(1-q)(V-\frac{s}{q})$, the difference between the two logarithms becomes

$$\log\left(\frac{V-\frac{s}{q}}{V}\frac{\underline{p}_1}{\underline{p}_1-\frac{s}{q}}\right)=\log\left(\frac{\underline{p}_1}{(1-q)V}\right).$$

The remaining term simplifies to $-\frac{\underline{p}_2}{s}$. Therefore,

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{\underline{p}_1}{(1-q)V}\right)-\frac{\underline{p}_2}{s}. \tag{29}$$

When both matches are good, expected seller revenues are

$$\mathbb{E}[\Pi_1\mid V,V]=\mathbb{E}\left[p_1\cdot\mathbf{1}_{\{p_1-p_2\le\frac{s}{q}\}}\right]=\frac{(1-q)V+s-(1-q)\mathbb{E}[p_1]}{q}, \tag{30}$$

$$\mathbb{E}[\Pi_2\mid V,V]=\mathbb{E}\left[p_2\cdot\mathbf{1}_{\{p_1-p_2>\frac{s}{q}\}}\right]=\frac{(1-q)\left(V-\frac{s}{q}\right)-(1-q)\mathbb{E}[p_2]}{q}. \tag{31}$$

These identities follow by conditioning on the seller's own price and substituting the other seller's CDF. With $(v_1,v_2)=(0,V)$, only seller 2 makes a sale and earns p_2. With $(V,0)$, only seller 1 makes a sale and earns p_1. With $(0,0)$, neither makes a sale. A bad-top-match buyer always searches at equilibrium prices, whereas a good-top-match buyer searches with the probability in (29). Applying the surplus identities in (16) and (17) gives Table 6.

Table 6: Conditional equilibrium outcomes and observed match frequencies

s	(v_1, v_2)	Count	$\mathbb{E}[\Pi_1]$	$\mathbb{E}[\Pi_2]$	$\mathbb{E}[\text{CS}]$	$\mathbb{E}[\text{TS}]$
20	$(0, 0)$	97	0.00	0.00	-20.00	-20.00
20	$(0, 100)$	119	0.00	49.93	30.07	80.00
20	$(100, 0)$	105	81.59	0.00	13.08	94.67
20	$(100, 100)$	99	58.41	10.07	26.19	94.67
40	$(0, 0)$	112	0.00	0.00	-40.00	-40.00
40	$(0, 100)$	104	0.00	18.96	41.04	60.00
40	$(100, 0)$	113	93.86	0.00	2.91	96.77
40	$(100, 100)$	121	86.14	1.04	9.60	96.77

Expectations are conditional on each match pair. Counts sum to 420 and 450. Platform revenue equals s in every row, top profit is top revenue minus s, and second profit equals second revenue. Treatment benchmarks weight the rows by their observed frequencies. Values are rounded for display.

D.3 Unconditional predictions

Let ρ denote the probability of search after a good top match in (29). A buyer with a bad top match always searches at equilibrium prices. Averaging over the initial match therefore gives

$$\mathbb{E}[x] = (1 - q) + q\rho. \tag{32}$$

At equilibrium, a product with value V is purchased whenever at least one match is good. The probability is $1 - (1 - q)^2 = q(2 - q)$. Hence

$$\mathbb{E}[\text{TS}] = q(2 - q)V - s[(1 - q) + q\rho], \tag{33}$$

$$\mathbb{E}[\text{CS}] = \mathbb{E}[\text{TS}] - \mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2]. \tag{34}$$

These formulas yield the unconditional values in Section 3.3. The empirical benchmarks instead weight the conditional outcomes in Table 6 by the observed match frequencies. The buyer's search decision in either calculation depends only on the initial match and posted prices, not the second match before search.

Appendix E Experimental instructions[5]

Welcome! Thanks for participating in this experiment on economic decision-making. Your participation is completely voluntary. Please remain seated and do not use your cellphone. You will interact anonymously through the computers. Please do not talk or look at others' screen.

At the end of the experiment, we will record your earnings amount and you will be paid by cash privately.

Fixed Role and Random Group

In this experiment, you will play the role of either a **Seller** or a **Buyer**. Your role will be assigned at the beginning and will remain fixed throughout the entire experiment.

There are **30 rounds** in total. In each round, you will be randomly grouped into a market consisting of **3 participants** (including yourself): **2 Sellers and 1 Buyer**. Groups are **randomly shuffled** every round, so you will likely interact with different people in the room from round to round. You will never know exactly who you are matched with.

Products and Fit Values

Sellers offer products to the Buyer. A seller's product can either be a Good Fit or a Bad Fit for the buyer, with equal chances.

- **Good Fit:** Value to the buyer is **100 points**. This happens with a chance of **50%**.
- **Bad Fit:** Value to the buyer is **0 points**. This happens with a chance of **50%**.

Before visiting a seller, the buyer does NOT know whether the product will be a good fit or bad fit. The fit is determined **randomly** and **independently** for each seller. The buyer only knows a product's fit value ***after visiting*** that seller. Knowing the fit with one seller does not provide any information about the fit with the other seller.

Decisions in Three Stages

Seller Bidding $\Longrightarrow$ Seller Pricing $\Longrightarrow$ Buyer Visit & Purchase

Each round proceeds according to the following three stages:

Stage I: Sellers Bid and Pay for Top Position

The two sellers simultaneously submit bids (in points) to compete for the "Top Position" on the Buyer's screen. A seller cannot see the other seller's bid when choosing their own bid. Bids must be between 0 to 100 points.

- **The Top Seller:** The seller with the higher bid wins and is displayed **first** to the buyer. It costs the buyer **0 points** to visit the **Top Seller**.
- **The Second Seller:** The seller with the lower bid loses and is displayed **second** to the buyer. It costs the buyer **20 points** to visit the **Second Seller**. The buyer decides whether or not to pay this cost, meaning the Second Seller might not be visited at all.

[5]These instructions are for the low-search-cost treatment. In the high-search-cost treatment, the search cost is 40 rather than 20 experimental points.

Tie-breaking: If both sellers submit the exact same bid, the computer will randomly select one seller to win and become the Top Seller (each seller has an equal chance).

How the Bidding Payment Works

The **Top Seller** wins the position, but pays a position payment equal to the **Second Seller's bid** (not their own bid). The **Second Seller** pays **0 points**.
Example: Seller A bids **22**. Seller B bids **14**. Seller A becomes the Top Seller, but Seller A only pays **14 points** (not 22). Seller B becomes the Second Seller and pays **0 points**.

Stage II: Sellers Set Prices Observed by Buyer

After sellers see whether they are the Top Seller or Second Seller, they simultaneously post a price for their product. Once posted, both prices are **immediately observed** by the buyer. When setting prices, sellers do not know if their product is a Good Fit or Bad Fit for the buyer.

Stage III: Buyer Visits and Purchases

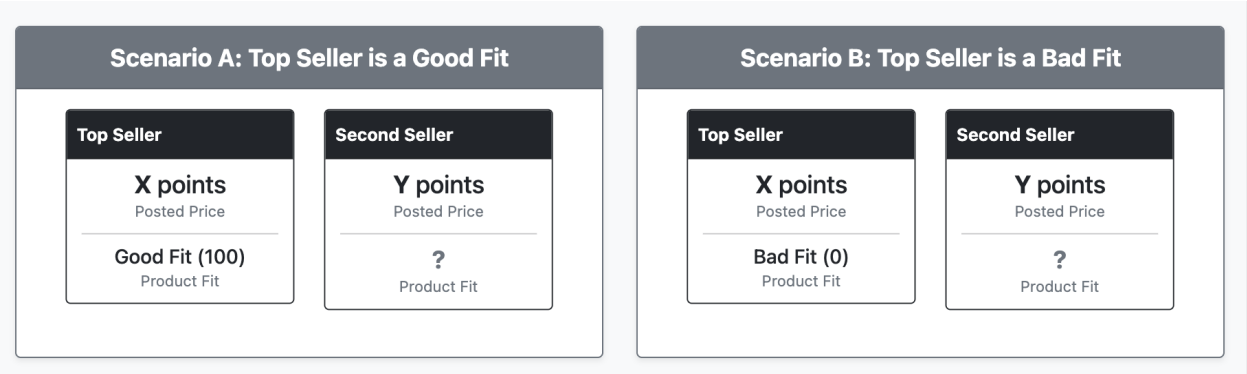
The buyer observes the prices posted by both sellers. The buyer automatically visits the **Top Seller** for free and learns its product fit.

The buyer then decides whether to pay a **Visit Cost** of **20 points** to discover the fit of the **Second Seller**. **The Visit Cost is non-refundable**, even if the buyer ultimately decides not to buy from the Second Seller after the visit.

Finally, the buyer can choose to purchase one product from any seller the buyer has visited. Specifically, a buyer who visits the Second Seller can still choose to buy from the Top Seller, without paying any *additional* visit cost. The buyer may also choose to quit the market at any time and buy nothing.

Examples of Buyer's Decision Screen

For example, if sellers post prices of **X** and **Y**, the buyer observes **one** of these two screens, depending on whether Top Seller is randomly determined to have a Good Fit or Bad Fit. Notice that buyer always knows Top Seller's fit, but Second Seller's fit remains **?** until visited.



Points and Earnings

At the end of the experiment, the computer will randomly select **ONE out of the 30 rounds** to be your "Payment Round." Your points in this Payment Round will be converted into dollar earnings using a conversion ratio of **10 points = $1**.

You will start each round with an initial balance of **200 points** (worth $20), which will be added to your payoff points in that round. This starting balance resets each round and does not carry over. Your total earnings for this experiment will be **$10** show-up fee plus your converted points in your (randomly selected) Payment Round.

Your payoff points are calculated as follows:

Buyer Payoff Points:

- If you purchase: **Product Fit** − **Price Paid** − **Visit Cost** (if you visit Second Seller)
- If you quit without purchase: − **Visit Cost** (if you visit Second Seller)

Notice that Visit Cost is non-refundable for the buyer.

Seller Payoff Points:

- If you make a sale: **Price Charged** − **Position Payment**
- If you do not make a sale: − **Position Payment**

Notice that Position Payment is non-refundable for sellers.

For each round: **Total Points = 200 Points + Payoff Points**

Please be patient while waiting for others to finish their decisions. At the end of each round, you will see a feedback screen showing your payoff and total points for that round.

Below is a summary of the instructions. Next, you will answer three comprehension questions and play two practice rounds. These practice rounds help you learn the interface and do not affect your earnings. After that, we will start the formal rounds. Once you have made a choice, you cannot go back to a previous screen to change it. **If you have any questions, please raise your hand at any time during the experiment.**

Instructions Summary

Rounds	**30 rounds**. Your role is fixed: you are either a **Seller** or a **Buyer**.
Groups	Each round: **2 Sellers + 1 Buyer** are **randomly** grouped.
Three Stages	Bidding $\Longrightarrow$ Pricing $\Longrightarrow$ Visit & Purchase
Bidding	Highest bid becomes **Top Seller** and pays the other Seller's bid.
Pricing	Prices posted by both Sellers are **observed** by Buyer before any visit.
Product Fit	When Buyer visits a Seller, Buyer discovers if the product is a **good fit** worth **100 points** (**50%** chance), or a **bad fit** worth **0 points** (**50%** chance). The fit is determined **independently** for each seller.
Visit Cost	Visiting **Top Seller** is free. Visiting **Second Seller** costs Buyer **20 points**.
$Earnings	Show-up **$10** + Points of **ONE** random round (**10 points = $1**).